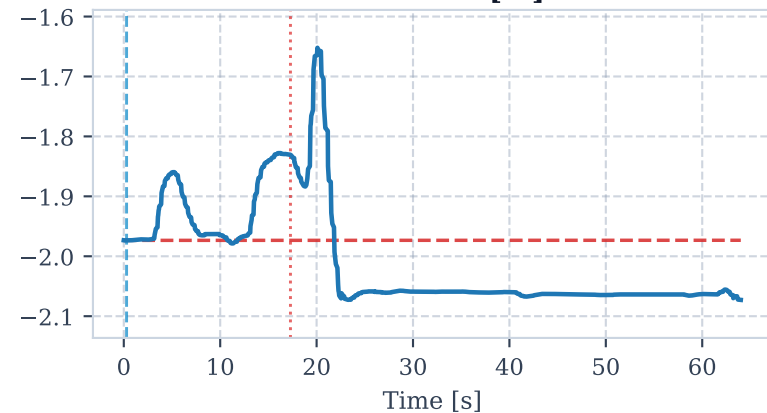


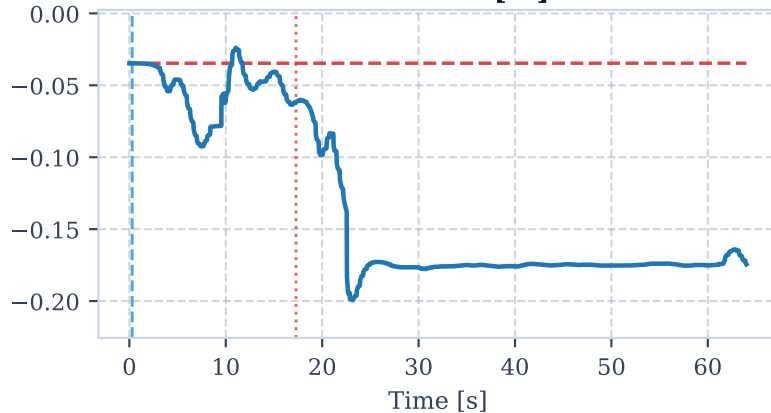
Drone Position vs MPC optimal trajectory



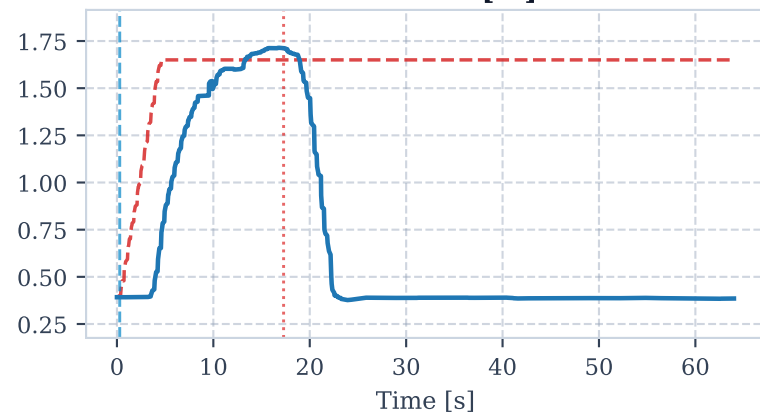
Position X [m]



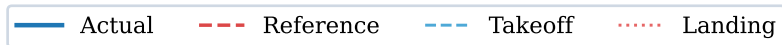
Position Y [m]



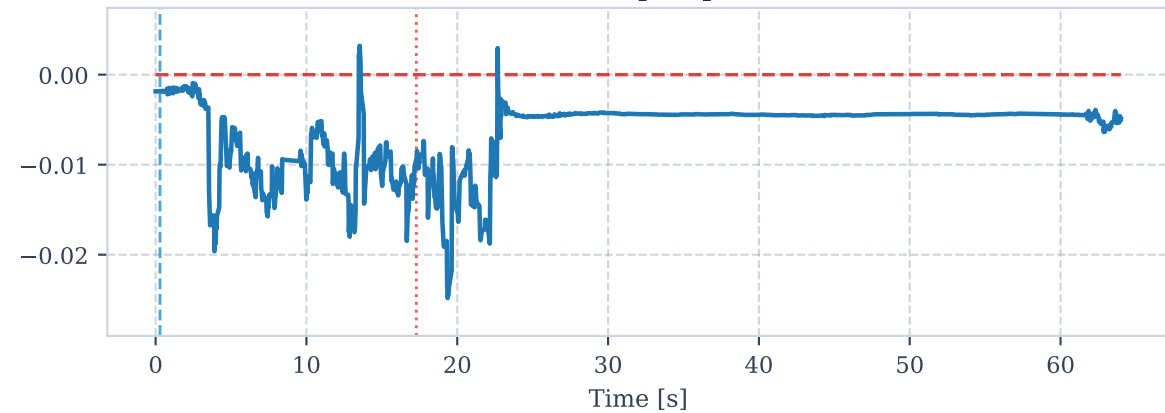
Position Z [m]



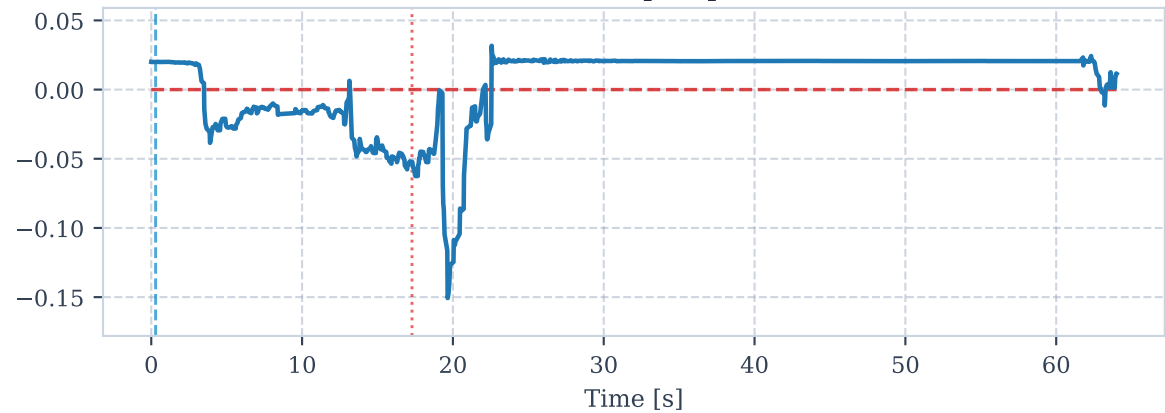
Drone Orientation (Roll/Pitch)



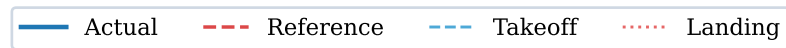
Roll [rad]



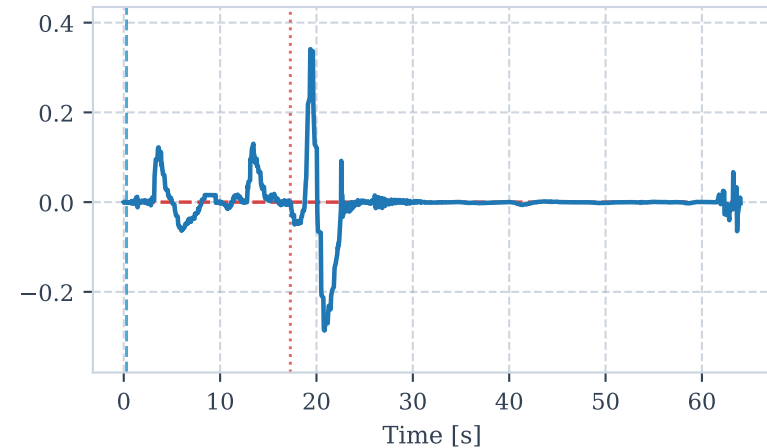
Pitch [rad]



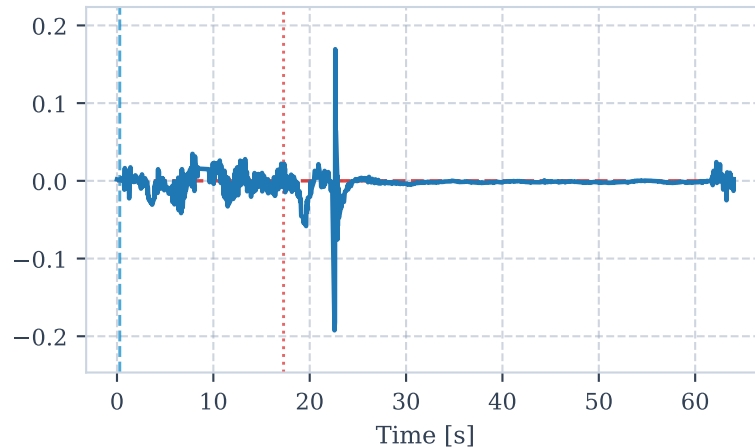
Drone Velocities vs MPC Reference



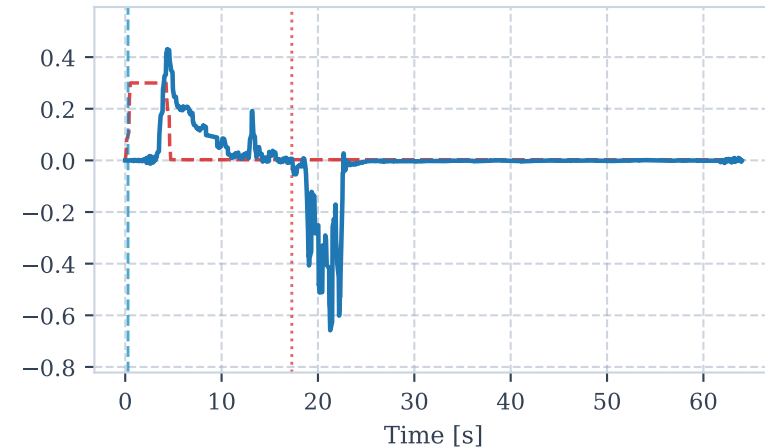
Vel X [m/s]



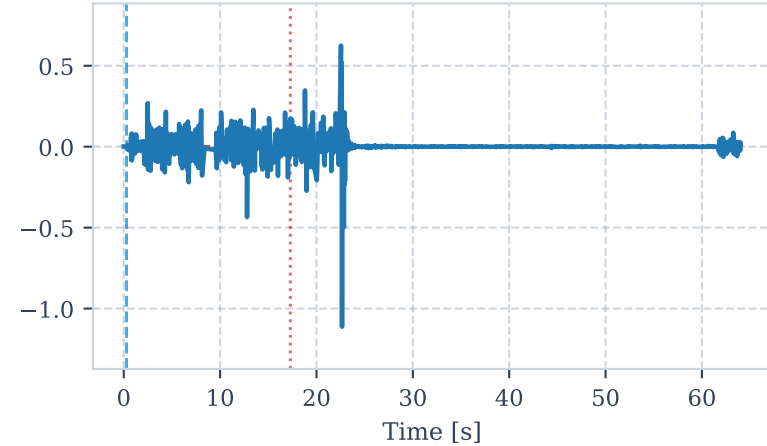
Vel Y [m/s]



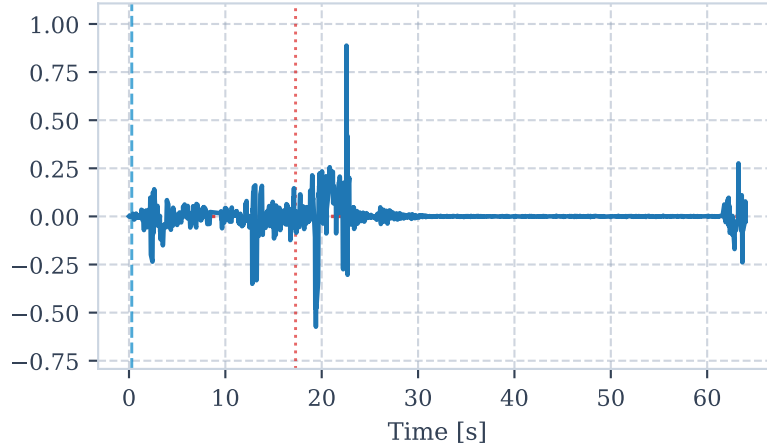
Vel Z [m/s]



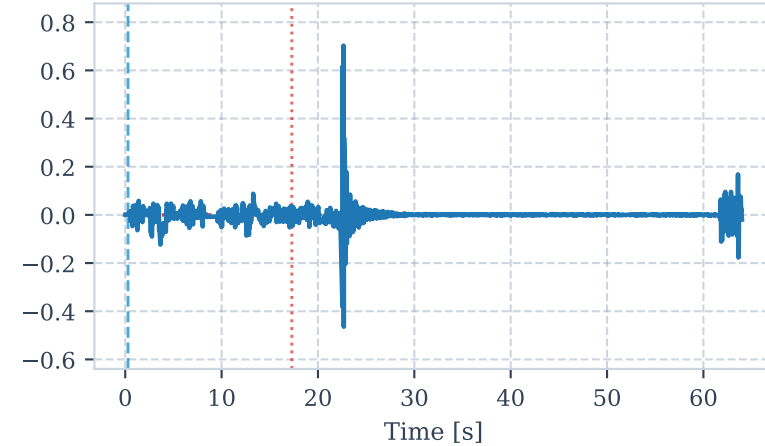
Omega X [rad/s]



Omega Y [rad/s]



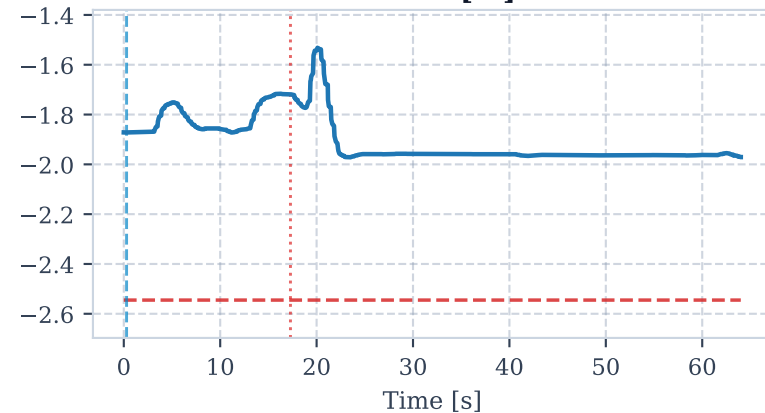
Omega Z [rad/s]



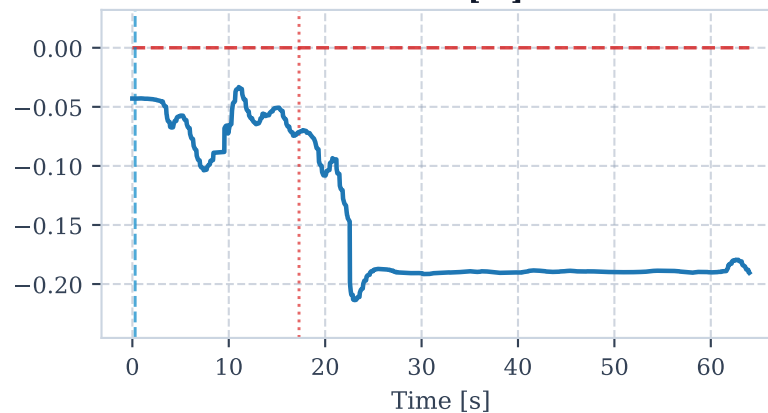
Camera Cartesian Position vs Derived Target



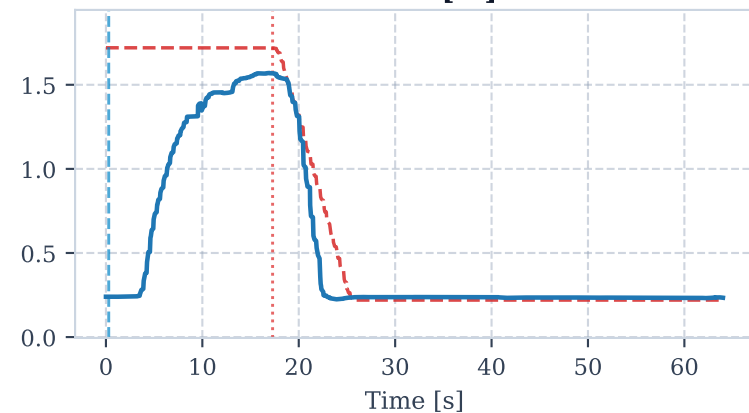
Cam X [m]



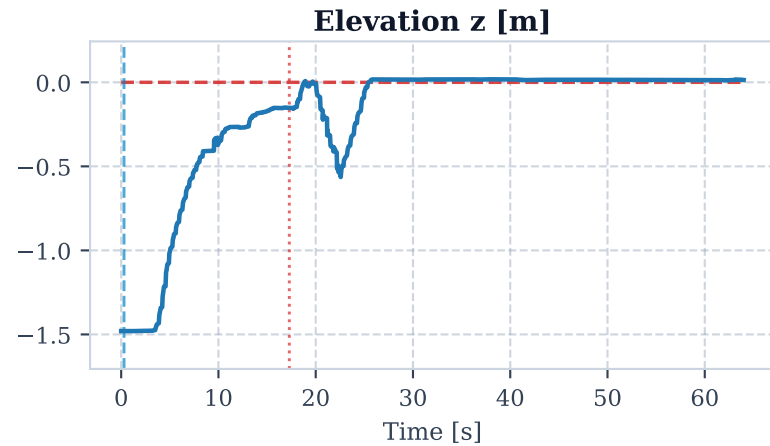
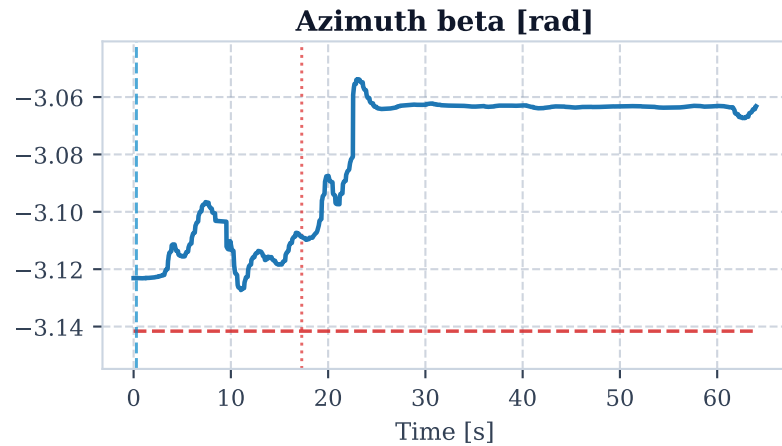
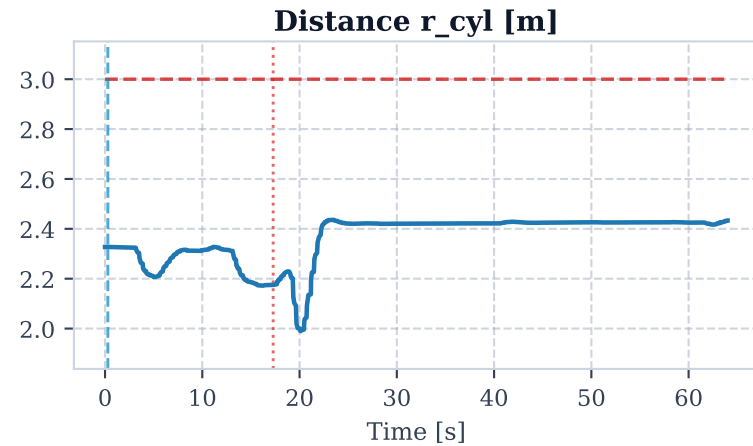
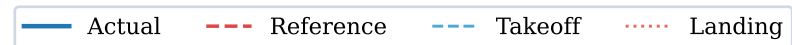
Cam Y [m]



Cam Z [m]



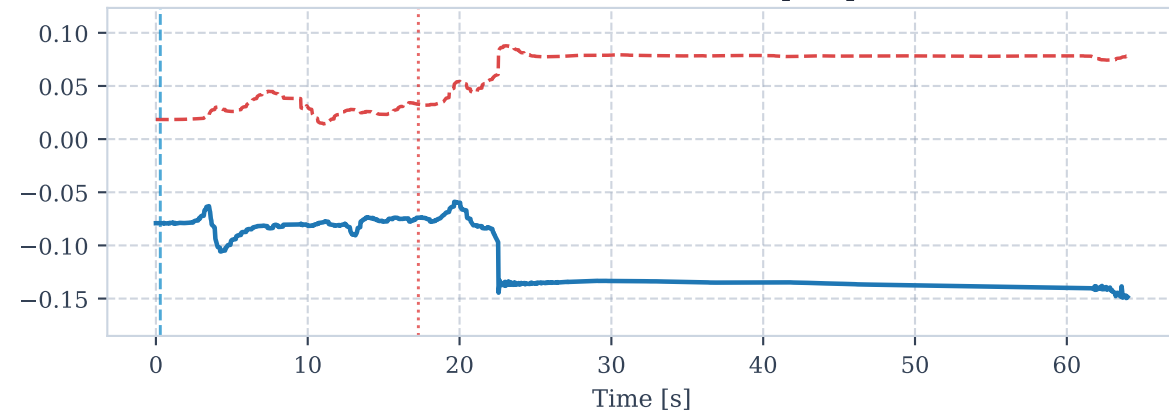
Cylindrical PoV Tracking (World Frame)



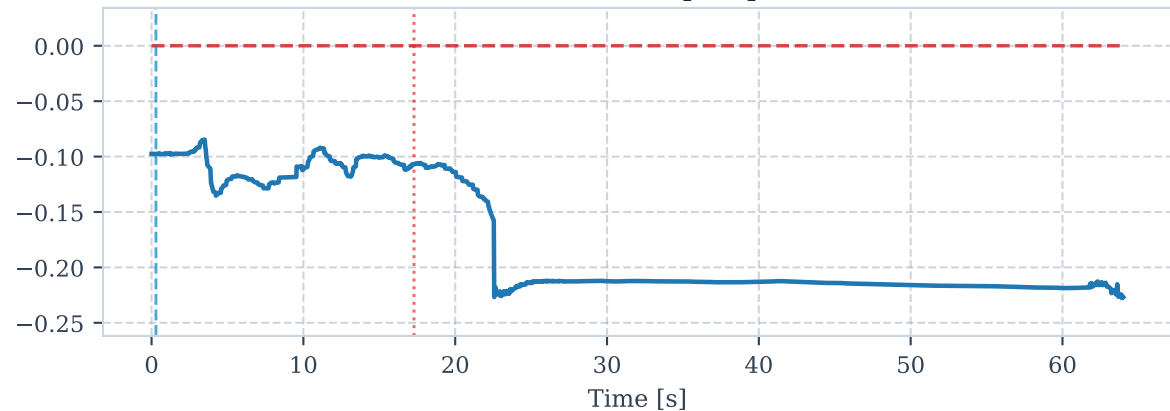
Yaw Tracking: Drone Pointing Target



Yaw Actual vs Desired [rad]



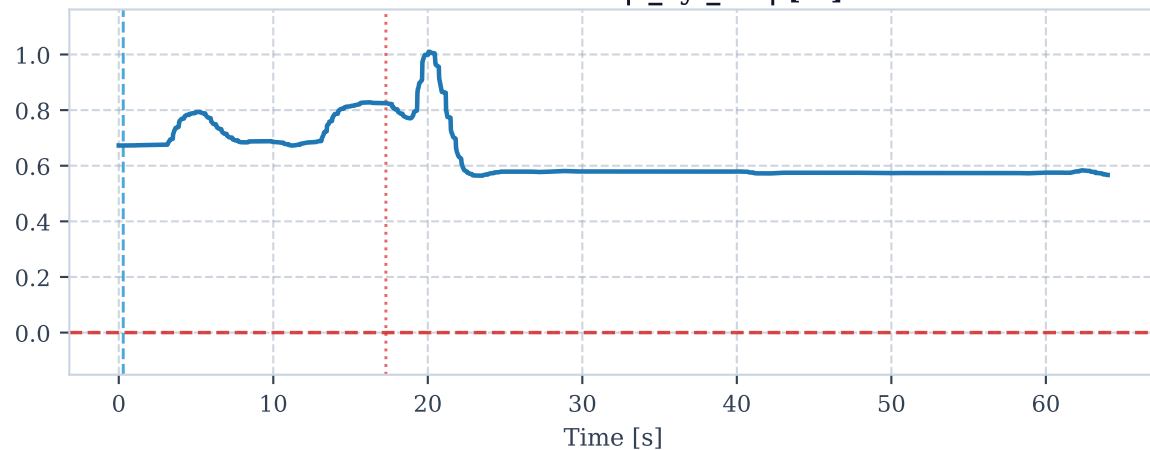
Yaw Error [rad]



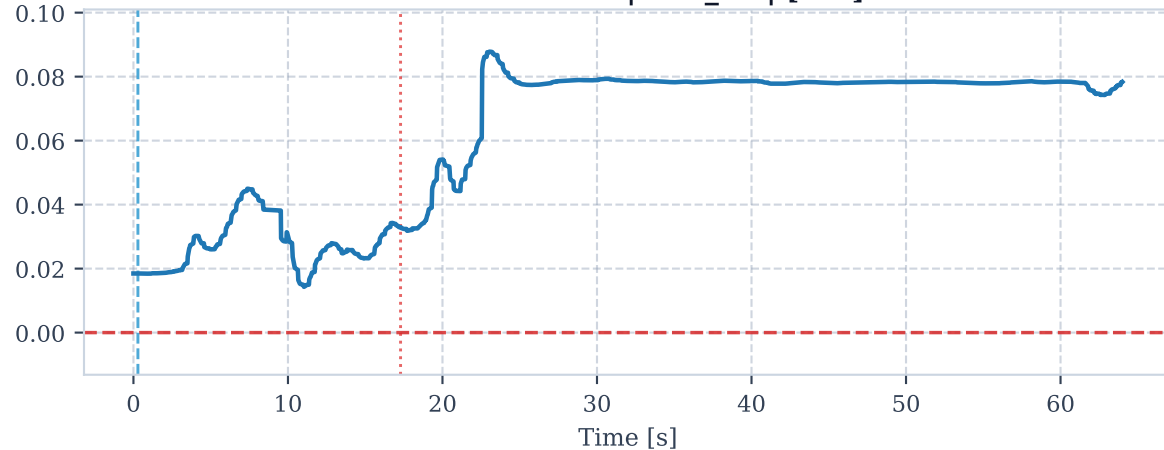
Primary Tracking Errors (Cylindrical & Yaw)

Actual Ref Takeoff Landing

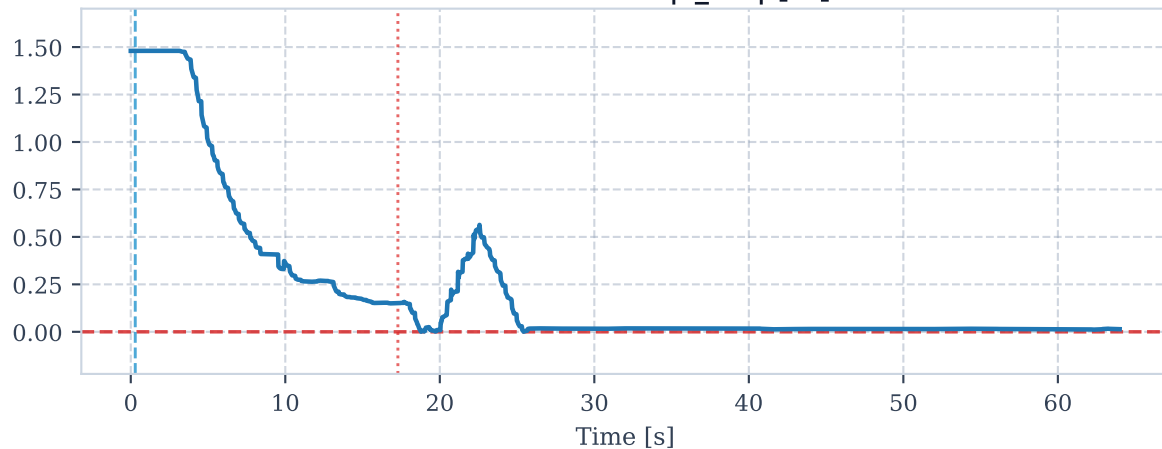
Distance Error $|r_{cyl_err}|$ [m]



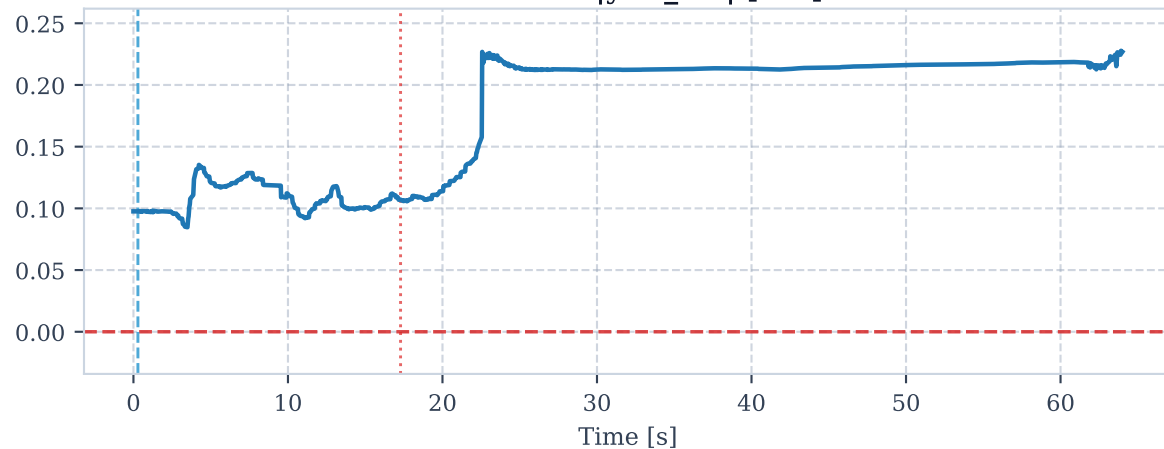
Azimuth Error $|\beta_{err}|$ [rad]



Elevation Error $|z_{err}|$ [m]

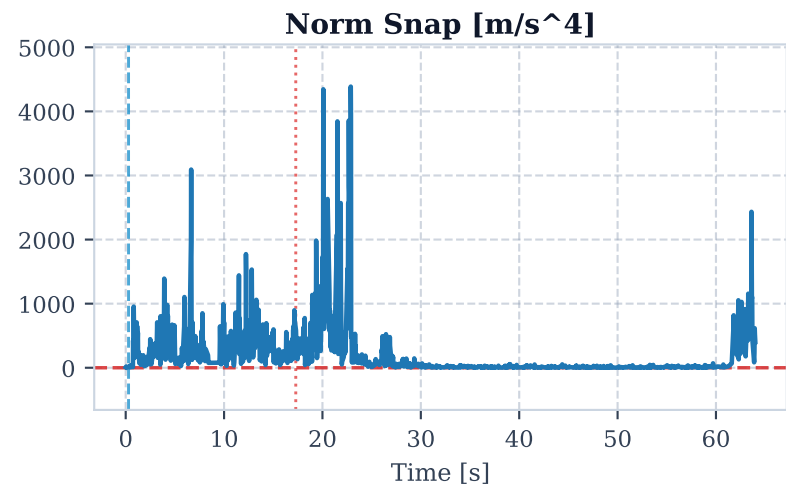
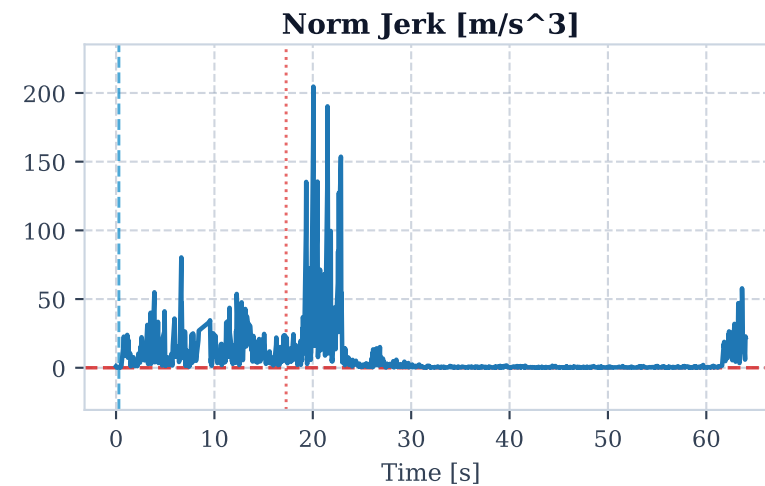
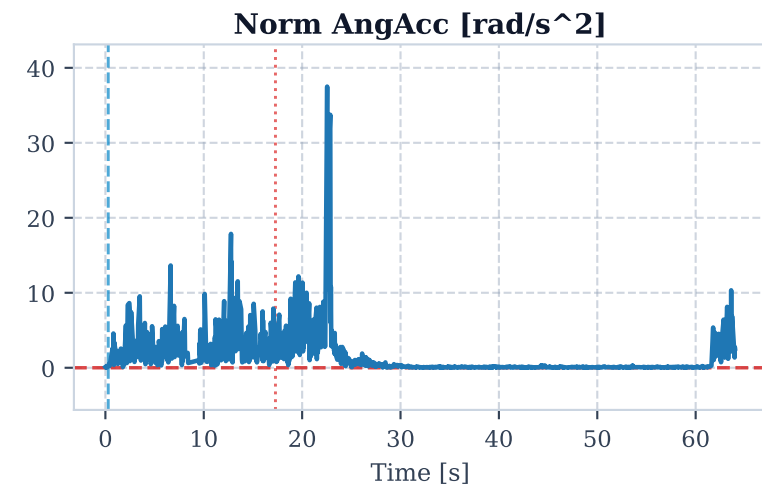
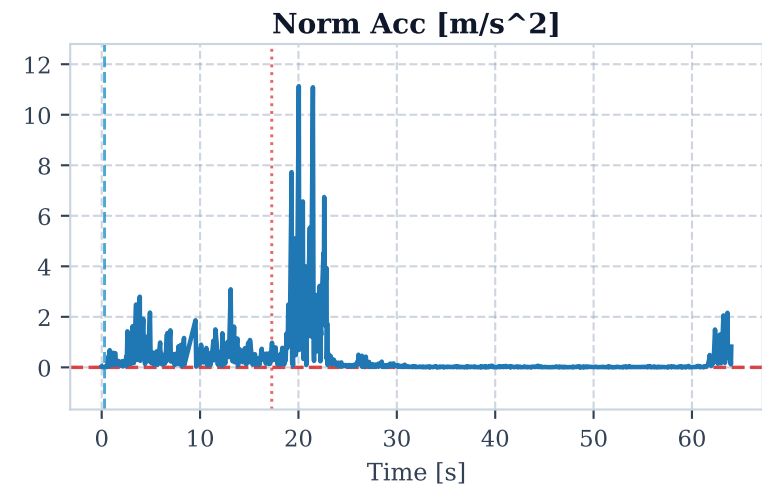
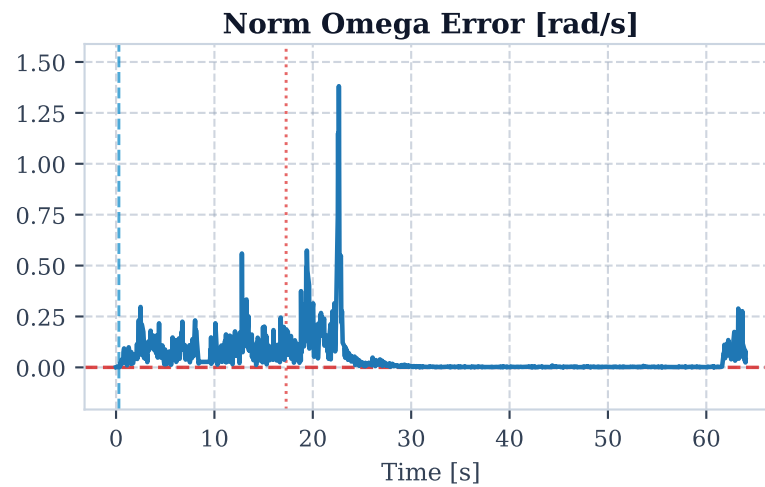
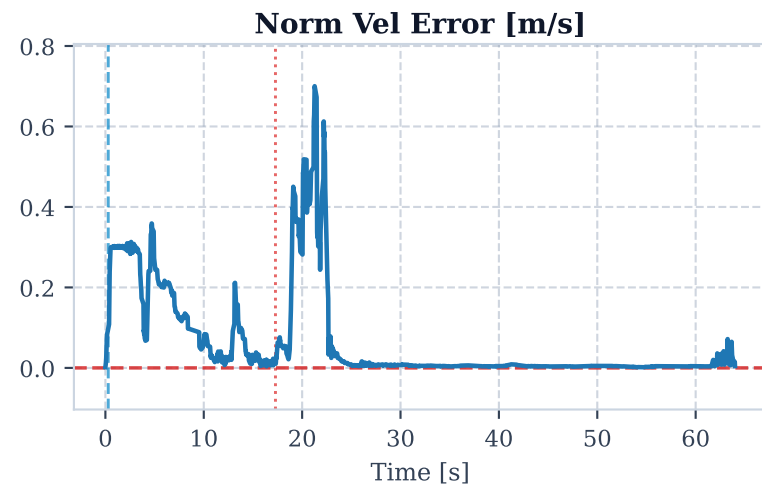


Yaw Error $|\text{yaw_err}|$ [rad]



Dynamic States Errors and Feedforward Derivatives

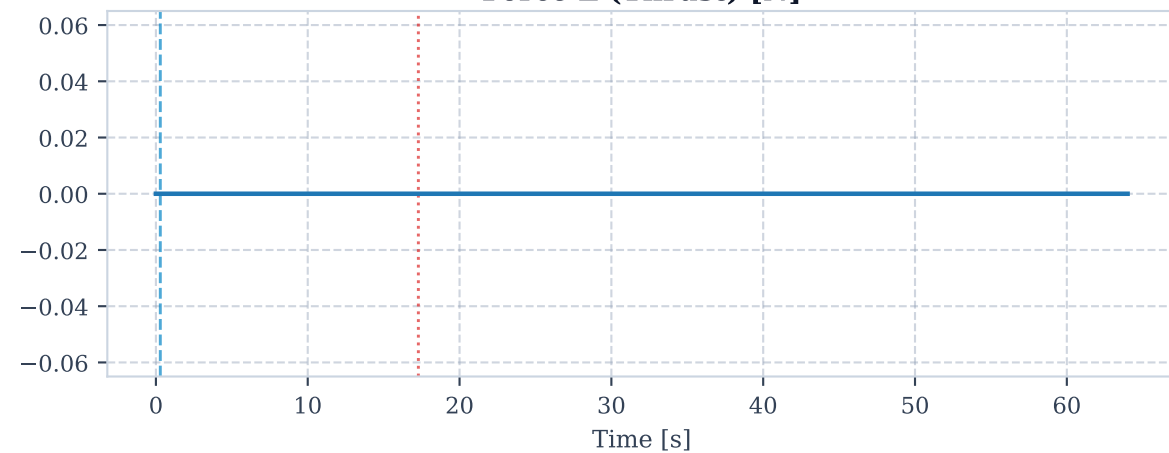
Actual Ref Takeoff Landing



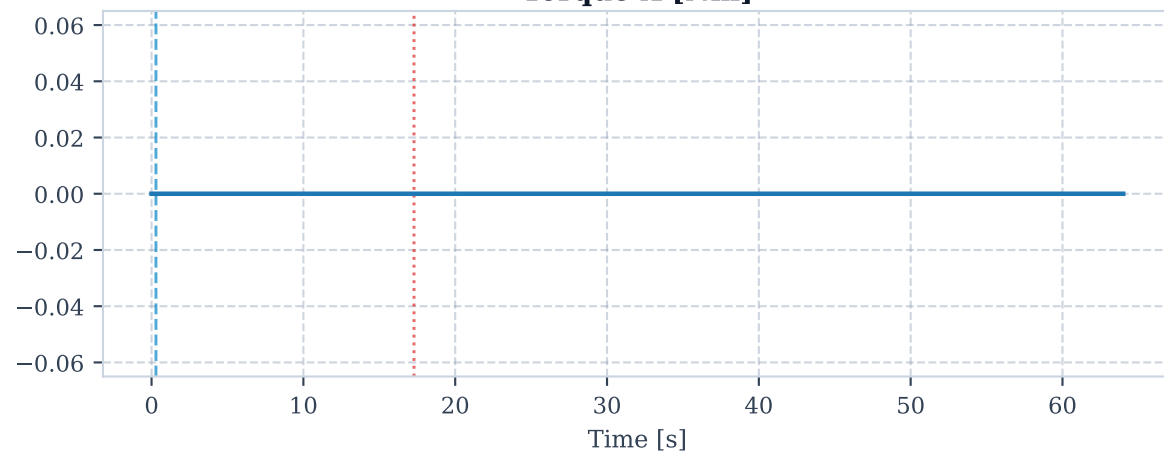
Control Wrench (Hover Force = 20.24N)

Actual Reference Takeoff Landing

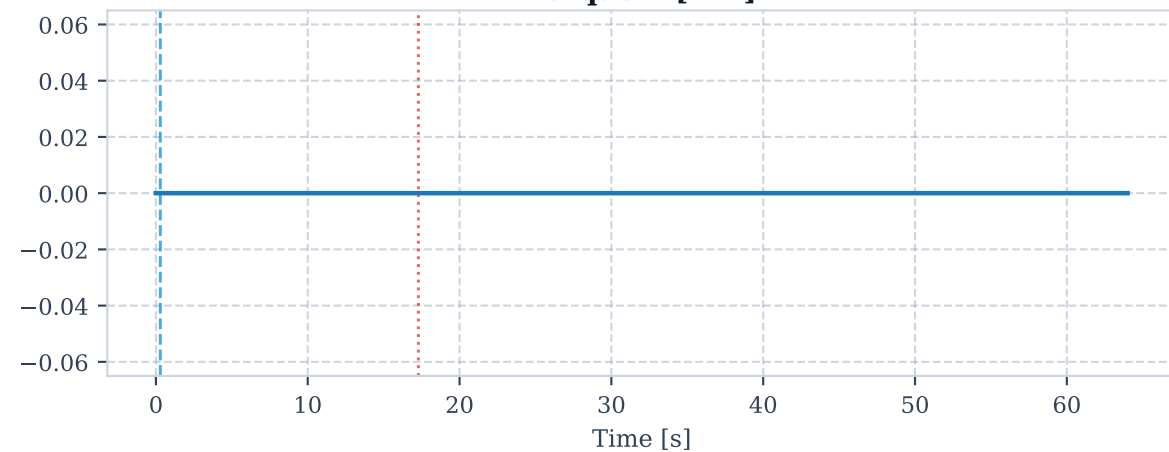
Force Z (Thrust) [N]



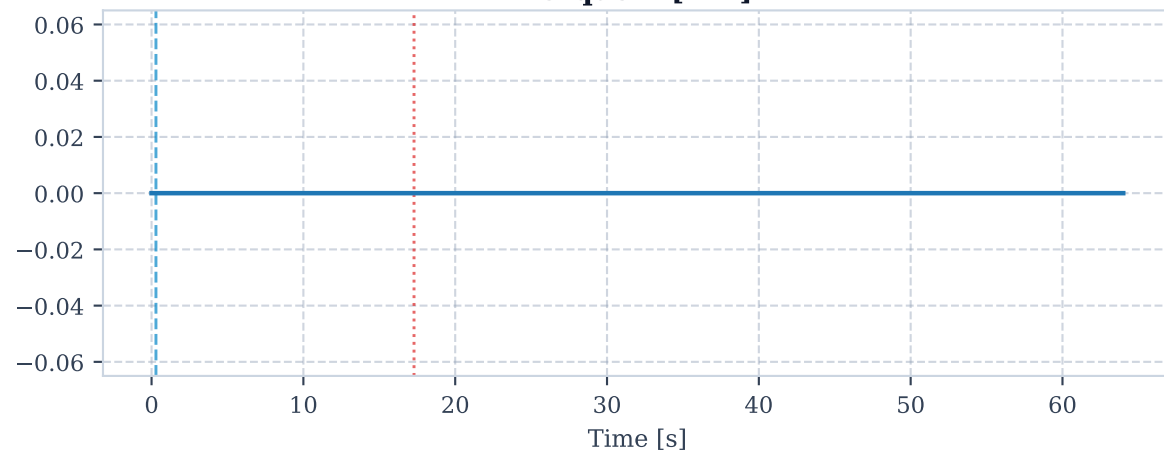
Torque X [Nm]



Torque Y [Nm]



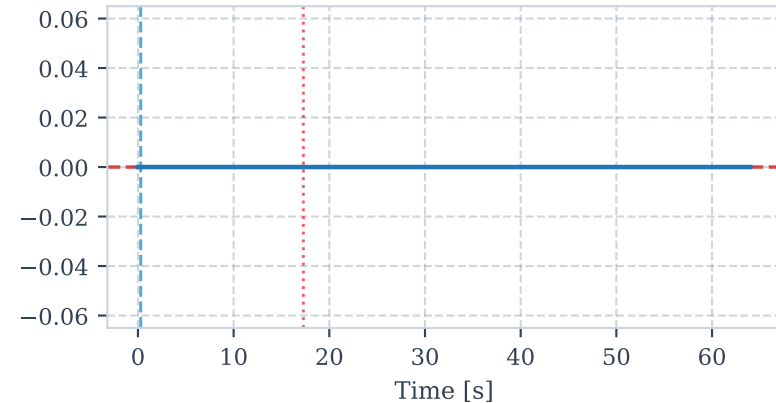
Torque Z [Nm]



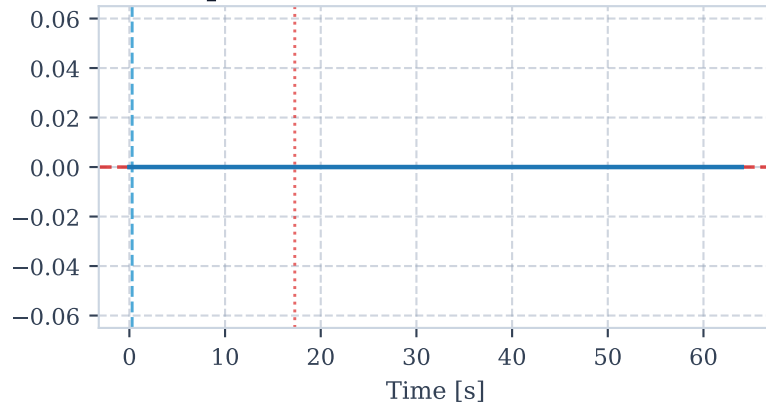
Haptic APF Repulsive Forces (Safety Constraints)



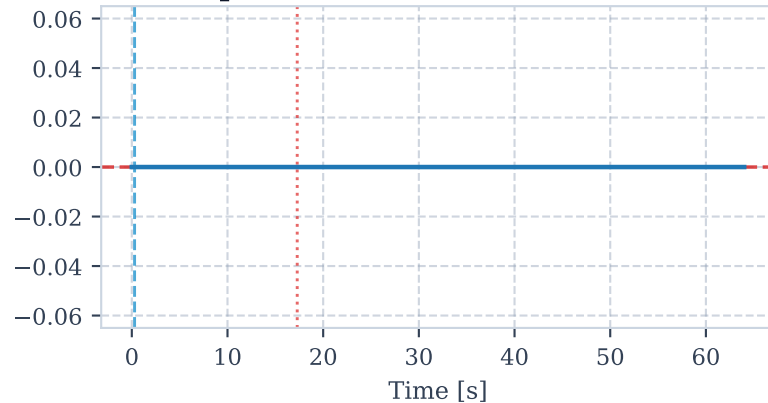
Repulsive Force X (Distance limit r_{min}) [N]



Repulsive Force Y (Horiz. FoV) [N]



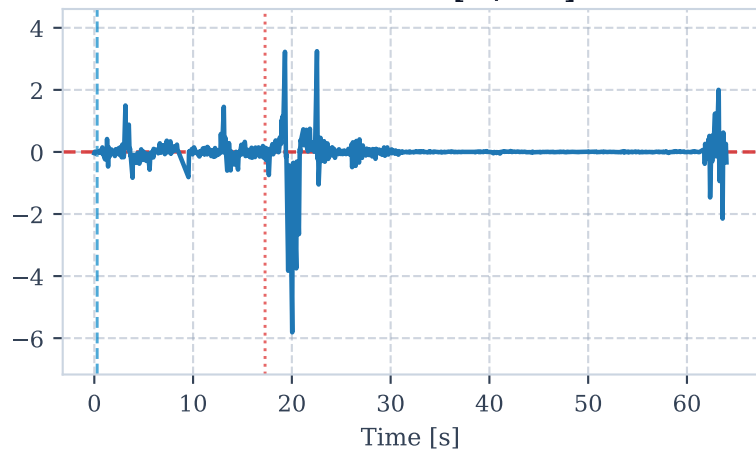
Repulsive Force Z (Vert. FoV) [N]



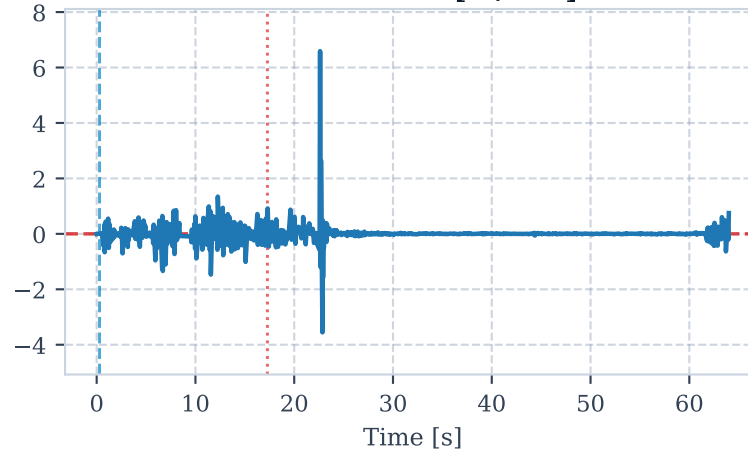
Drone Linear and Angular Accelerations

Actual Ref Takeoff Landing

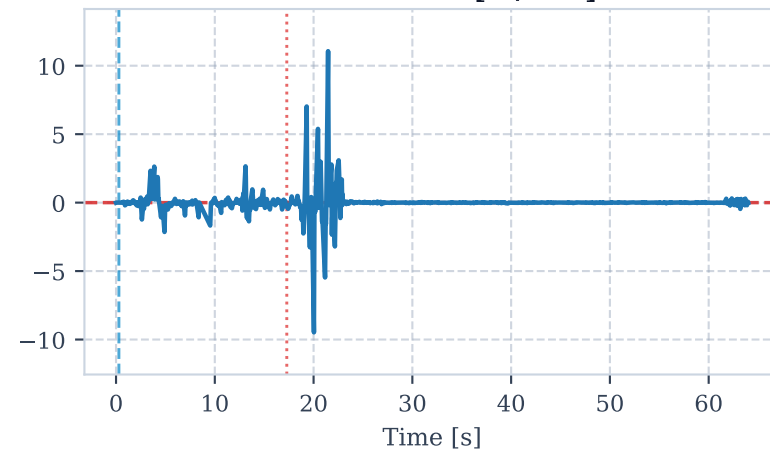
Linear Acc X [m/s²]



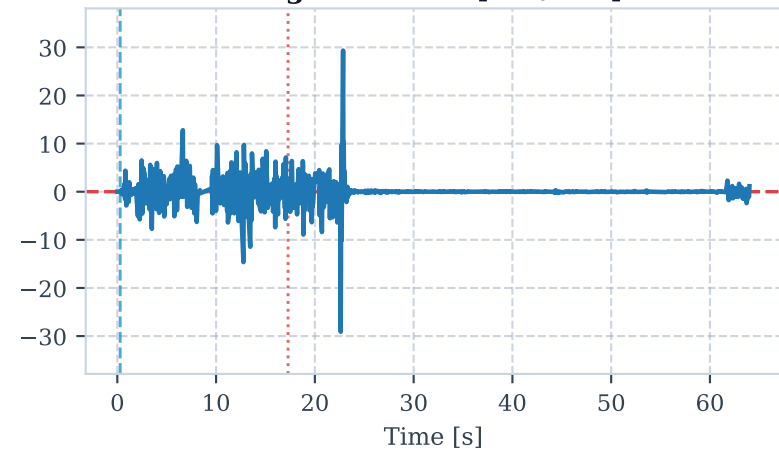
Linear Acc Y [m/s²]



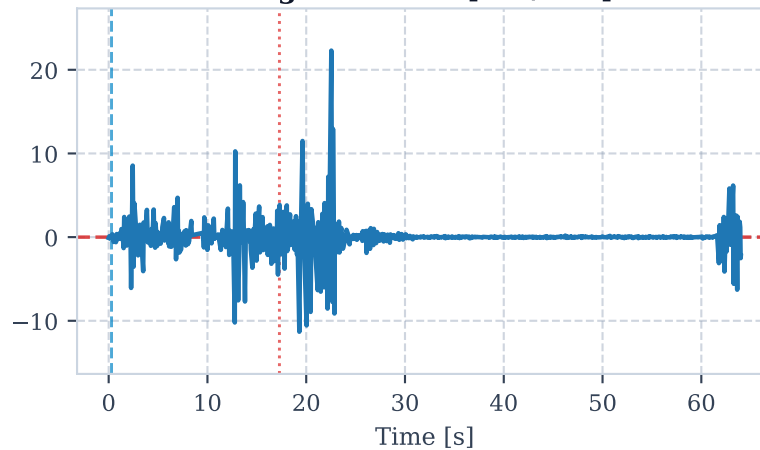
Linear Acc Z [m/s²]



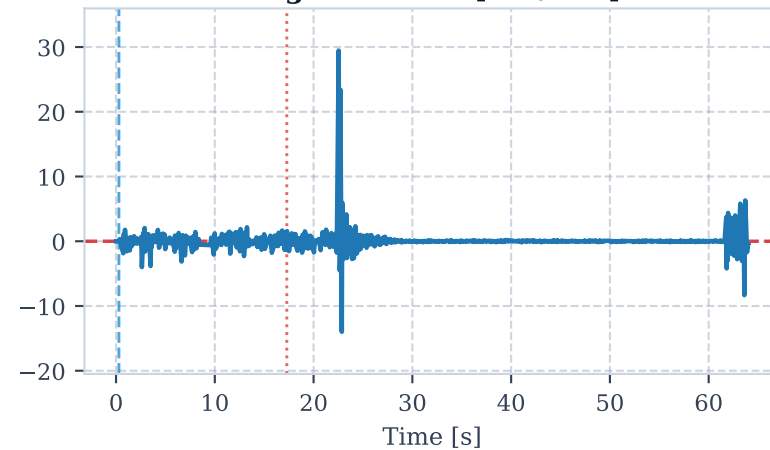
Angular Acc X [rad/s²]



Angular Acc Y [rad/s²]



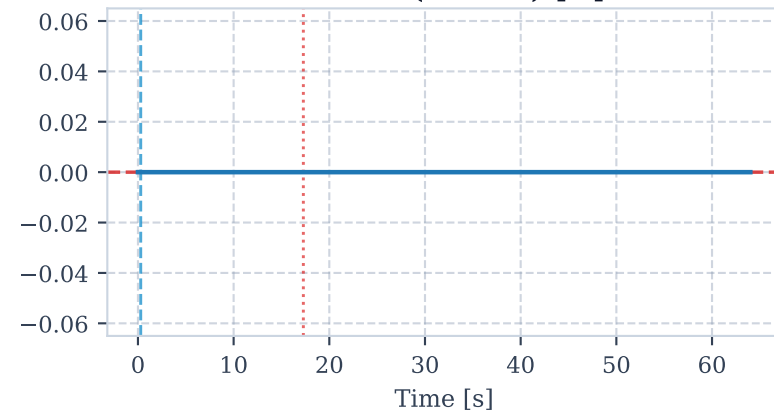
Angular Acc Z [rad/s²]



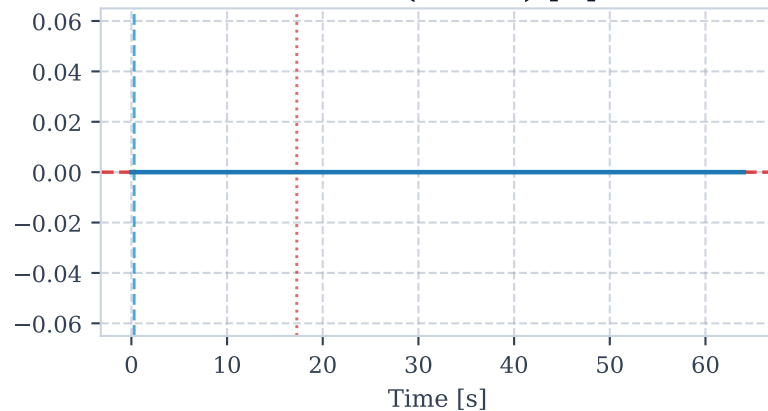
Peg External Contact Forces (FT Sensor)



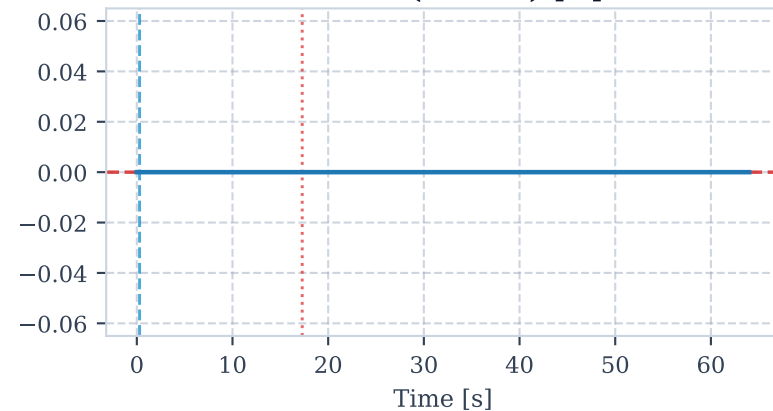
Force X (Sensor) [N]



Force Y (Sensor) [N]

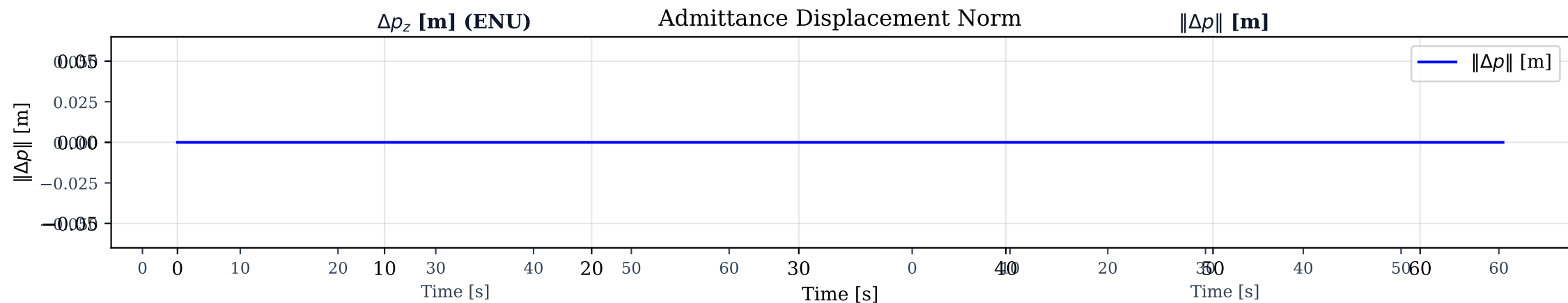
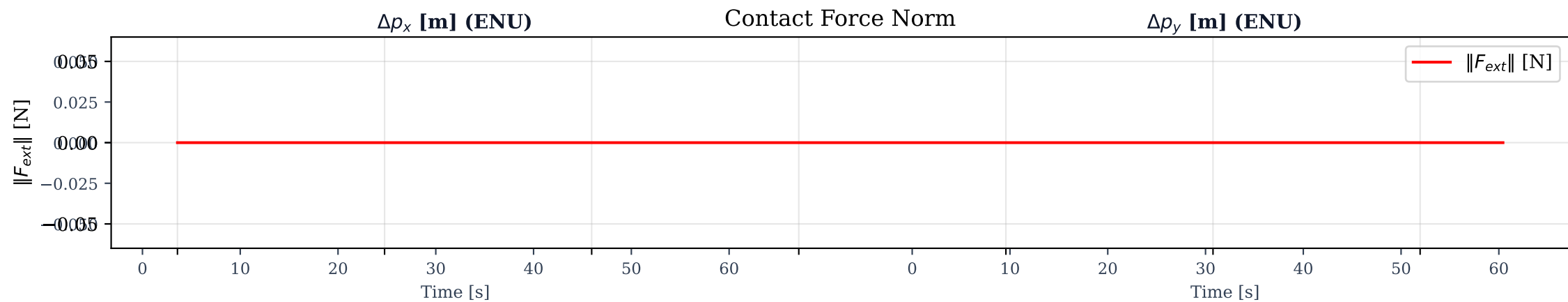


Force Z (Sensor) [N]

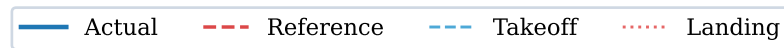


Admittance Effect: Contact Force vs Position Deviation

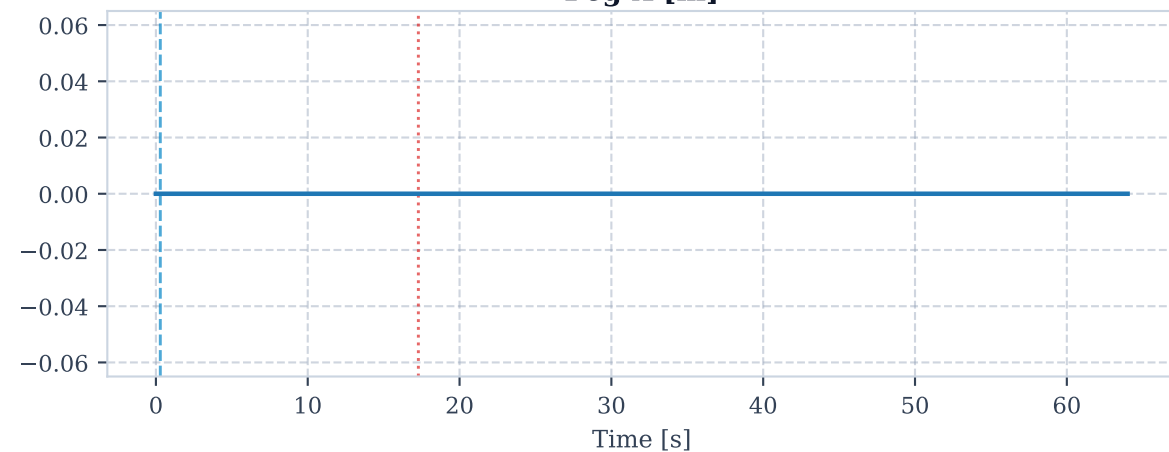
Actual Ref Takeoff Landing



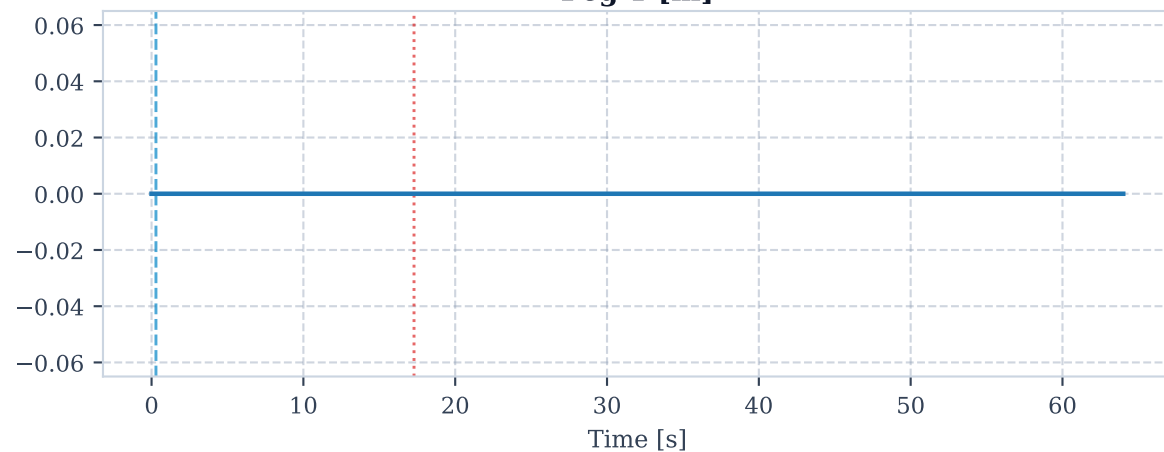
Interaction Drone Position ENU (Actual vs Planner Reference)



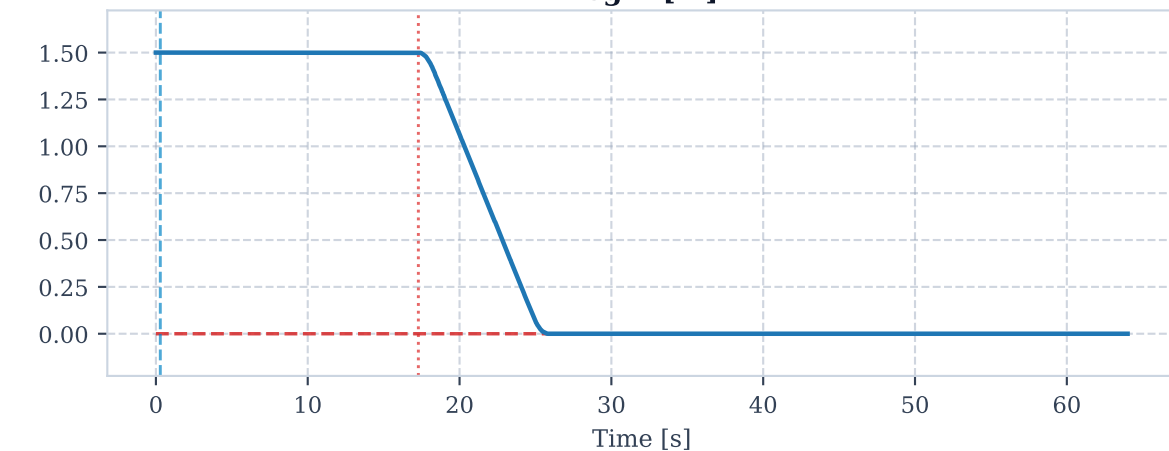
Peg X [m]



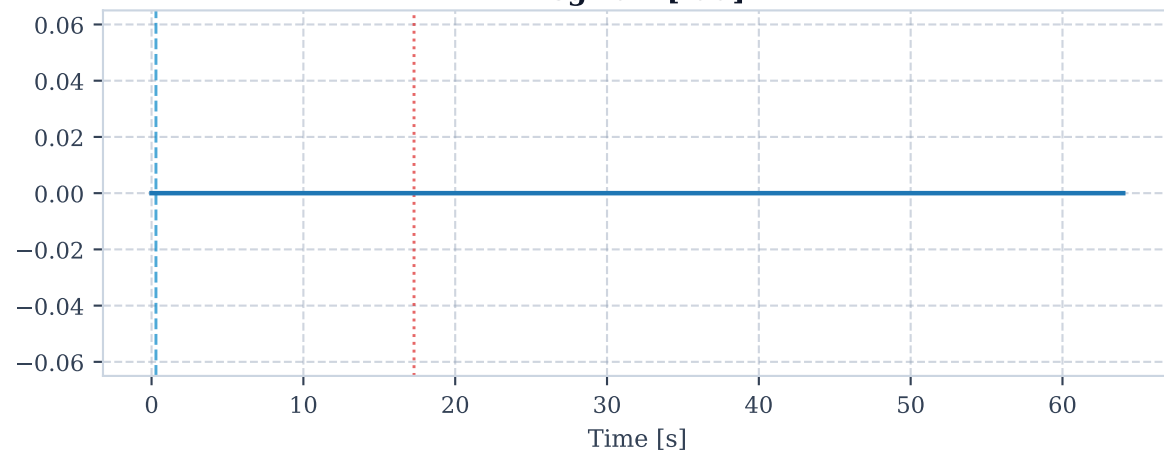
Peg Y [m]



Peg Z [m]



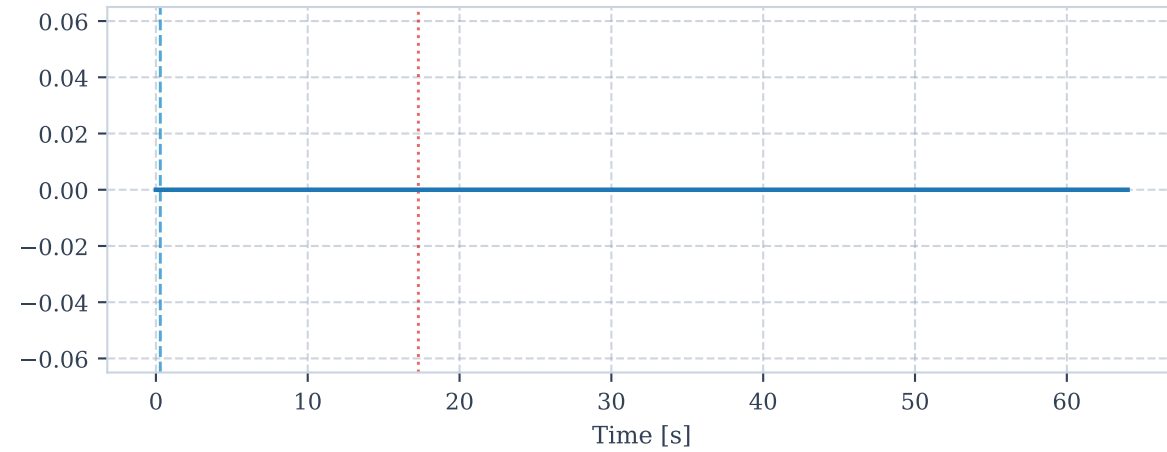
Peg Yaw [rad]



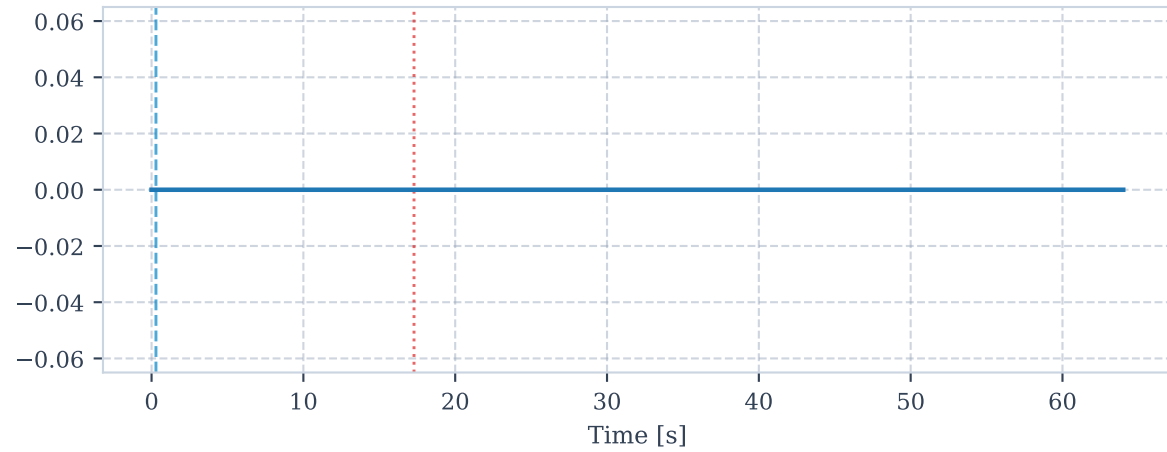
Interaction Drone Velocities (ENU) and Yaw Rate

— Actual - - - Reference - - - Takeoff ····· Landing

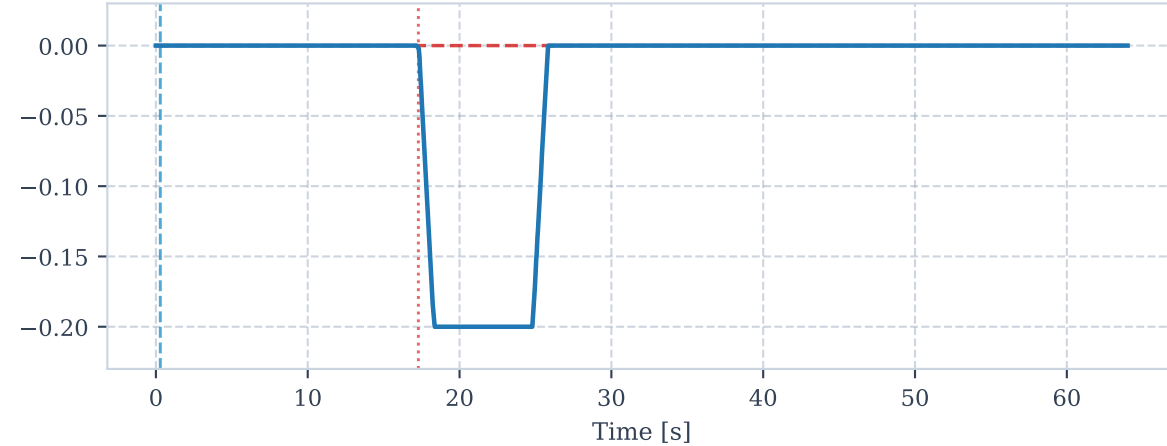
Vel X [m/s]



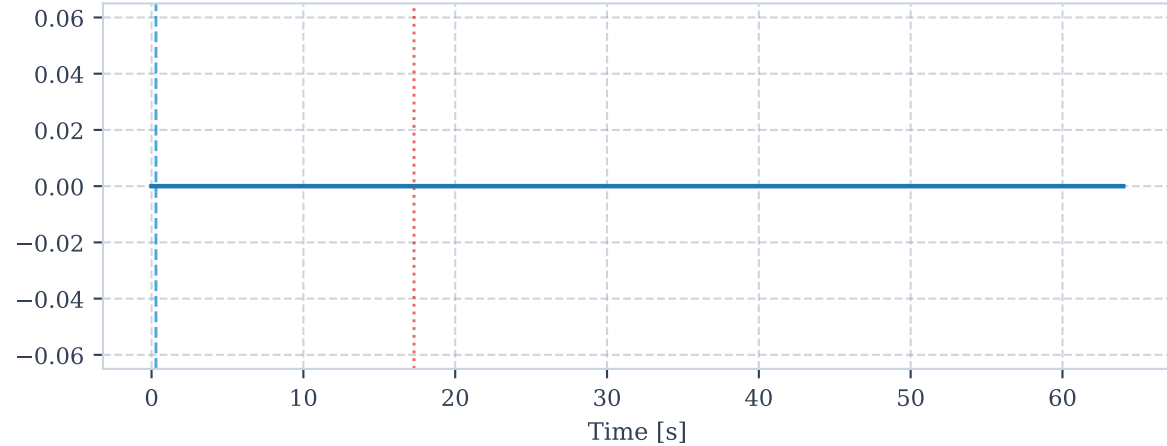
Vel Y [m/s]



Vel Z [m/s]



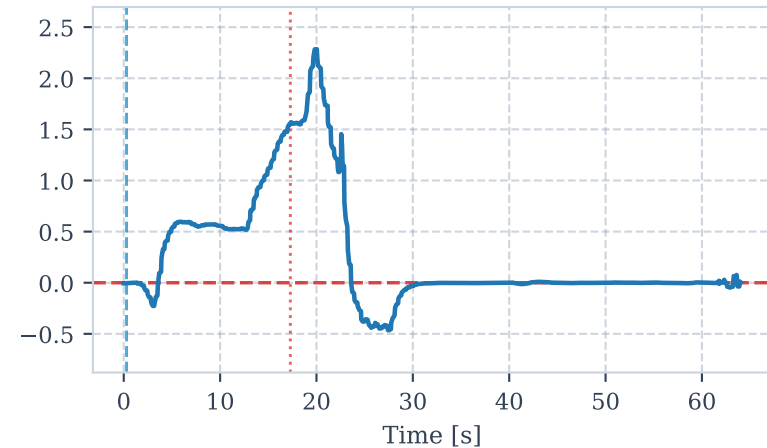
Yaw Rate [rad/s]



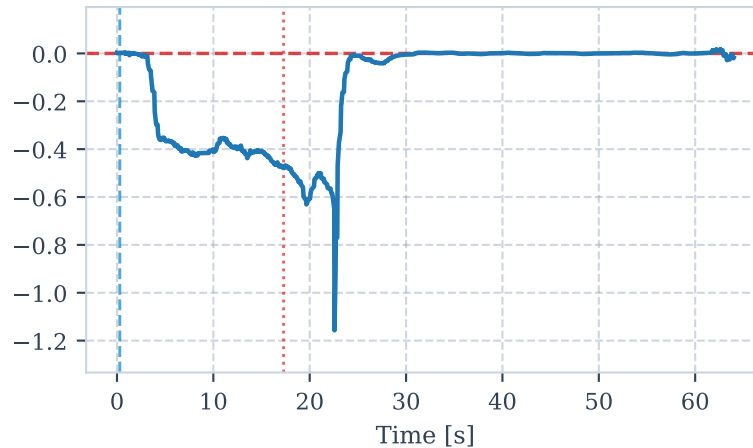
Estimated Wrench (Momentum-Based Estimator)

Actual Ref Takeoff Landing

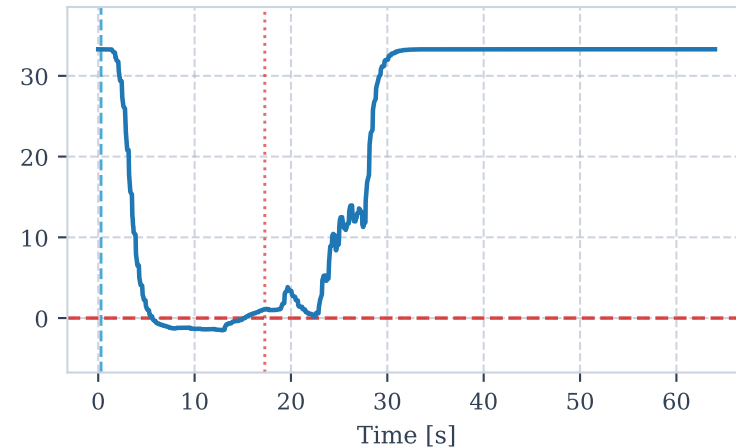
Force X [N]



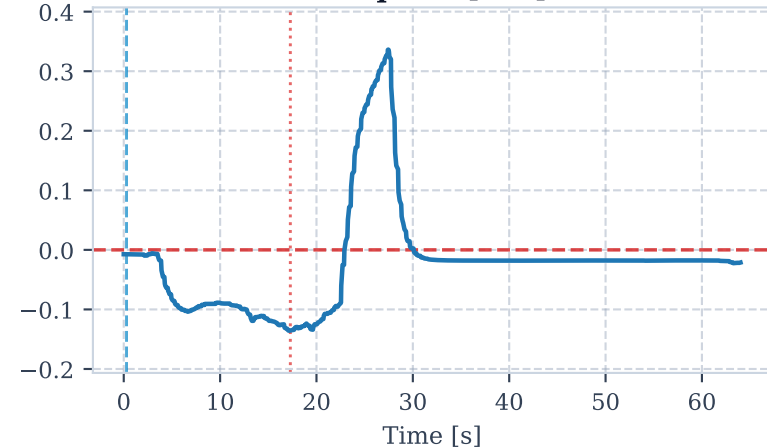
Force Y [N]



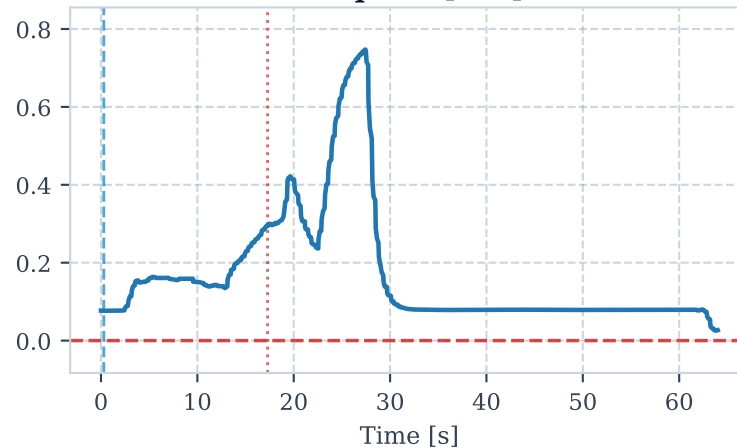
Force Z [N]



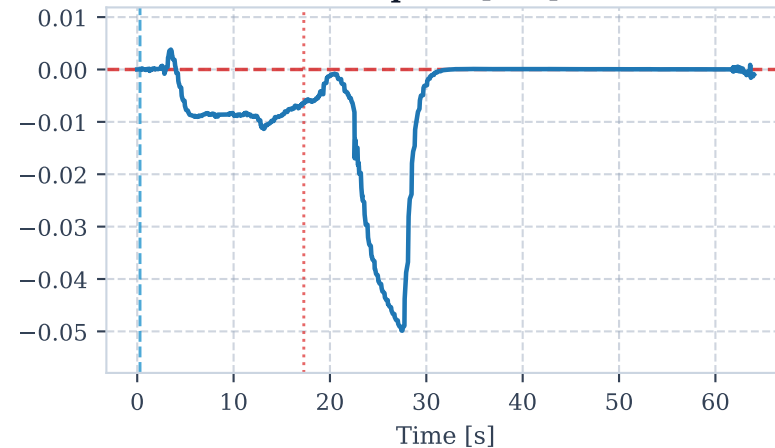
Torque X [Nm]



Torque Y [Nm]



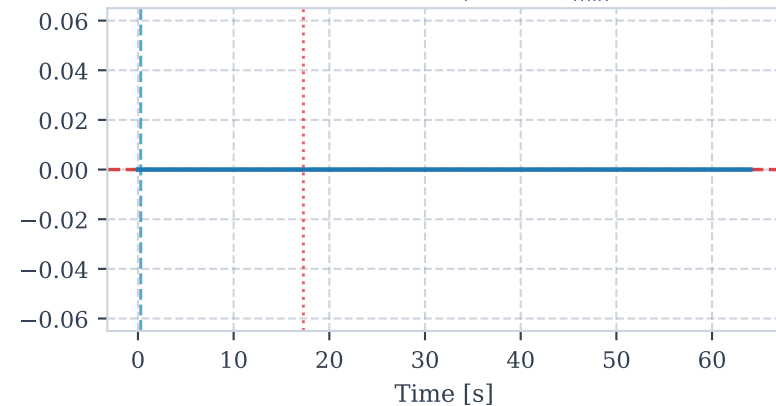
Torque Z [Nm]



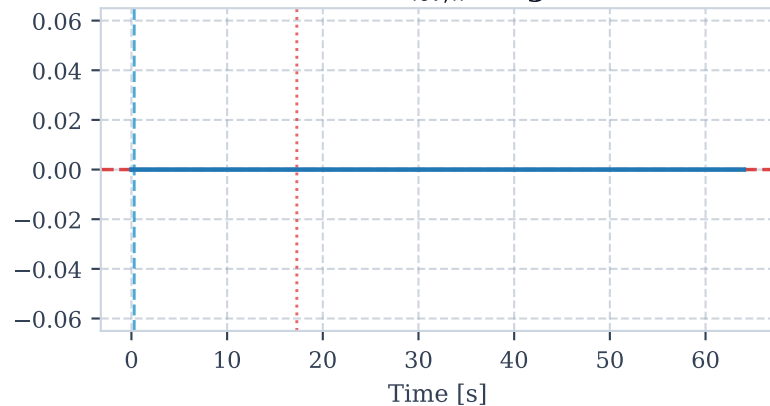
Safety Distance and Camera FoV Constraint Violations



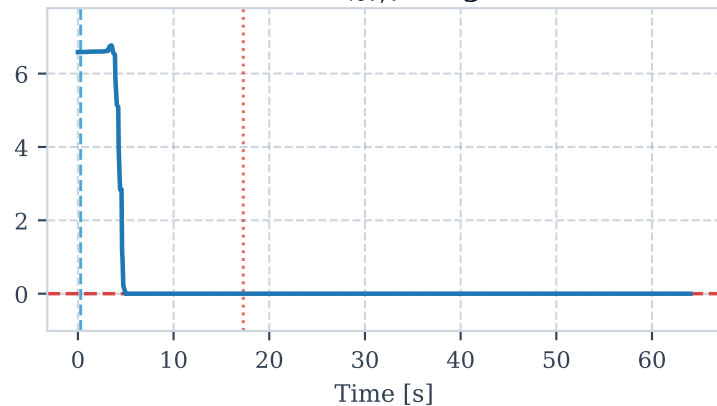
Distance Violation s_r [m] ($r_{min}=1.0$ m)



Horiz. FoV Violation $s_{fov,h}$ [deg] (half-FoV=34°)



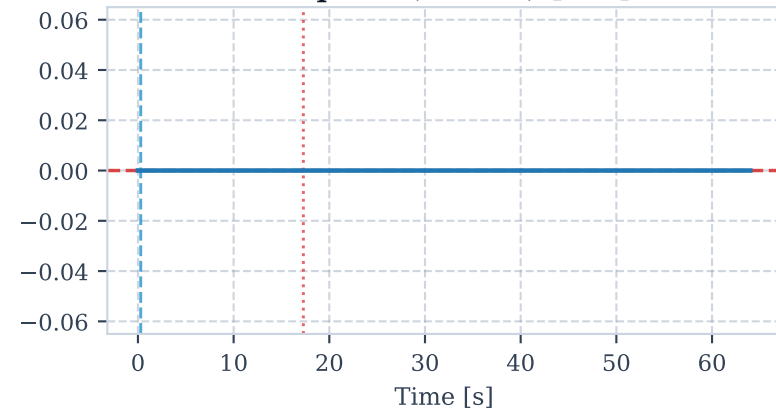
Vert. FoV Violation $s_{fov,v}$ [deg] (half-FoV=26°)



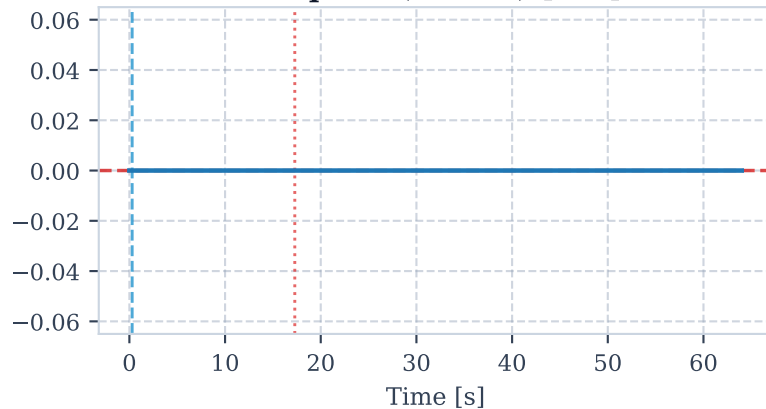
Peg External Contact Torques (FT Sensor)



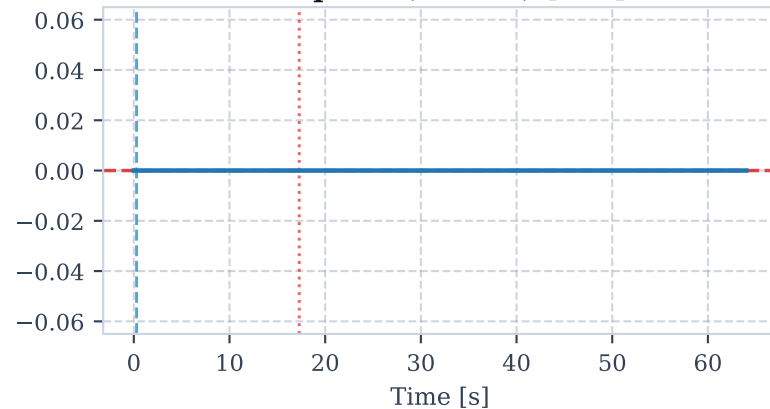
Torque X (Sensor) [Nm]



Torque Y (Sensor) [Nm]



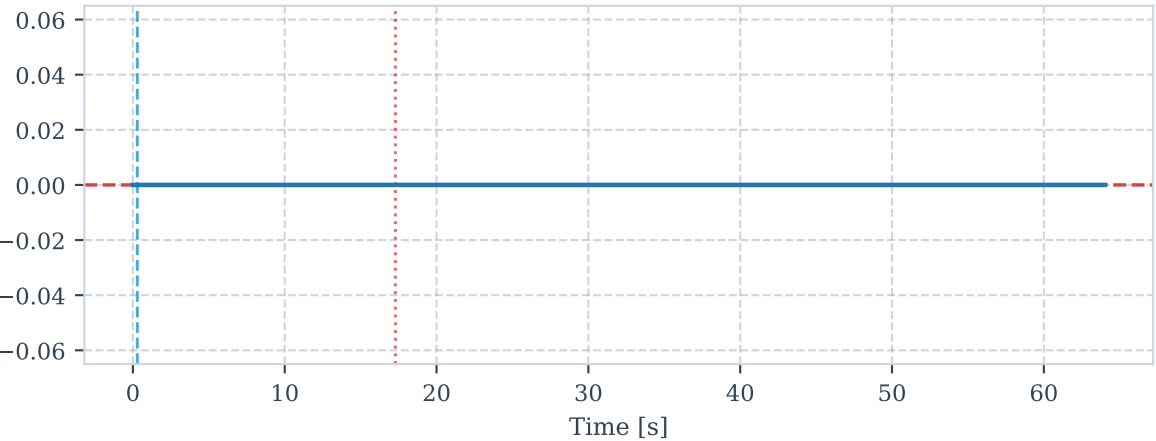
Torque Z (Sensor) [Nm]



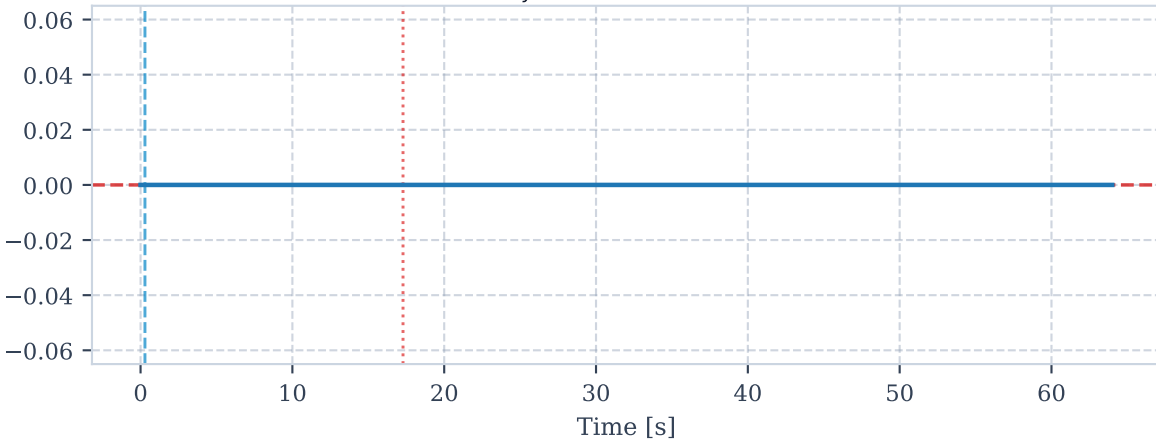
Admittance Displacement in Sensor Frame



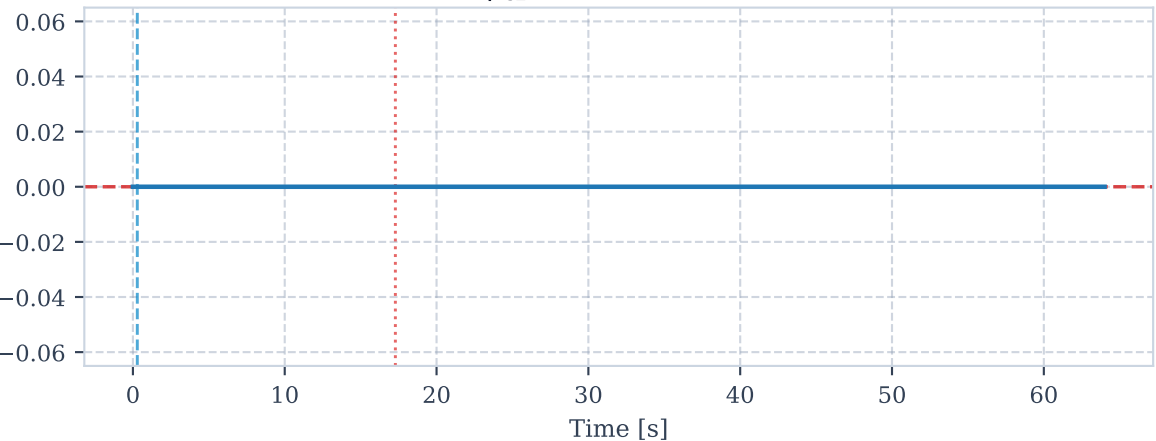
Δp_{sx} [m] (Sensor X)



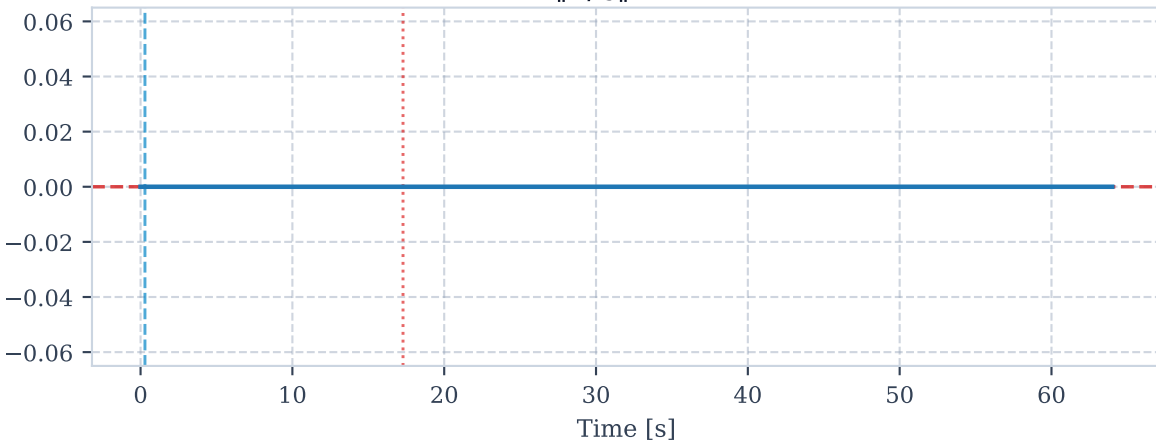
Δp_{sy} [m] (Sensor Y)



Δp_{sz} [m] (Sensor Z)



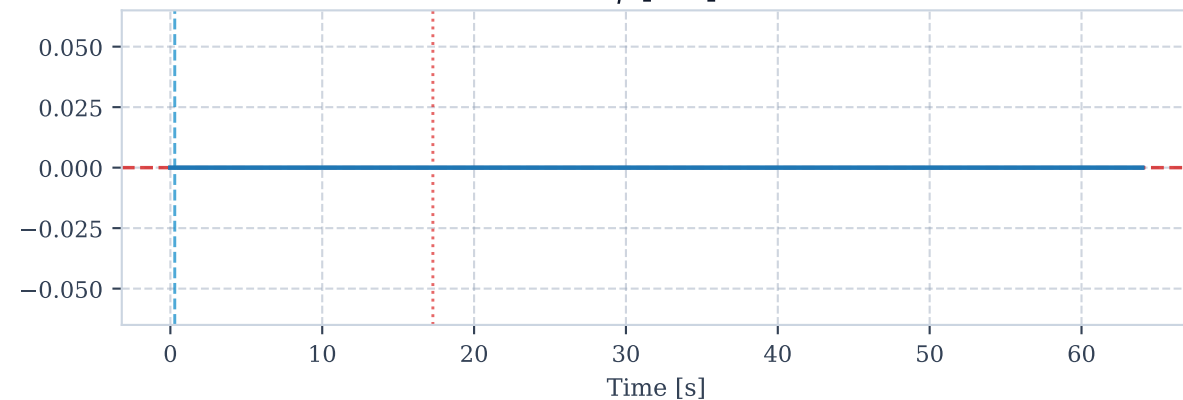
$\|\Delta p_s\|$ [m]



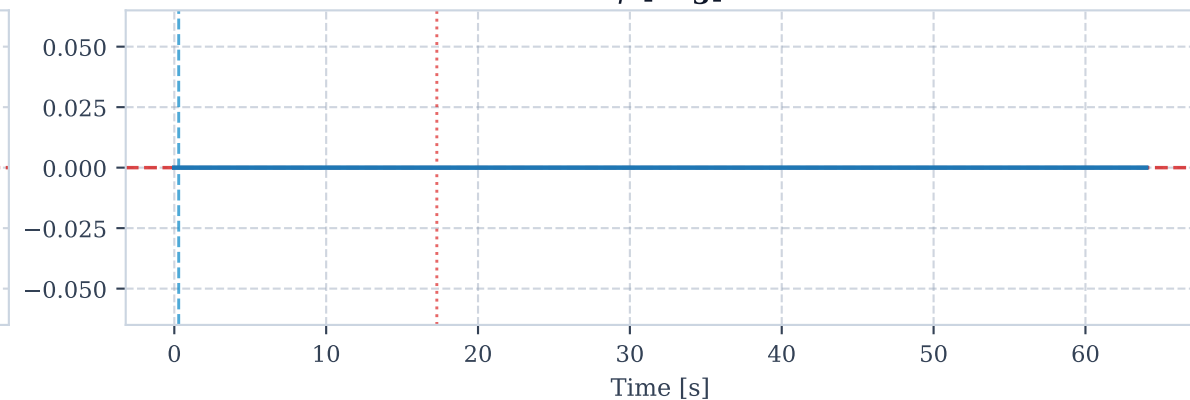
Admittance Yaw Displacement $\Delta\psi$



$\Delta\psi$ [rad]



$\Delta\psi$ [deg]



Rotational Admittance Effect: Contact Torque τ_x vs $\Delta\psi$

